

Diagram illustrating the decay chains of various isotopes, showing the sequence of decays (alpha, beta, gamma) and the resulting daughter products. The diagram is organized into a grid with columns representing the mass number (A) and rows representing the atomic number (Z).

The isotopes shown include:

- Actinides (Z=89-95):** ²⁰⁹Ac, ²¹⁰Ac, ²¹¹Ac, ²¹²Ac, ²¹³Ac, ²¹⁴Ac, ²¹⁵Ac, ²¹⁶Ac, ²¹⁷Ac, ²¹⁸Ac, ²¹⁹Ac, ²²⁰Ac, ²²¹Ac, ²²²Ac, ²²³Ac, ²²⁴Ac, ²²⁵Ac, ²²⁶Ac, ²²⁷Ac, ²²⁸Ac, ²²⁹Ac, ²³⁰Ac, ²³¹Ac, ²³²Ac.
- Thorium (Z=90):** ²¹³Th, ²¹⁴Th, ²¹⁵Th, ²¹⁶Th, ²¹⁷Th, ²¹⁸Th, ²¹⁹Th, ²²⁰Th, ²²¹Th, ²²²Th, ²²³Th, ²²⁴Th, ²²⁵Th, ²²⁶Th, ²²⁷Th, ²²⁸Th, ²²⁹Th, ²³⁰Th, ²³¹Th, ²³²Th, ²³³Th, ²³⁴Th, ²³⁵Th, ²³⁶Th.
- Protactinium (Z=91):** ²¹⁶Pa, ²¹⁷Pa, ²¹⁸Pa, ²¹⁹Pa, ²²⁰Pa, ²²¹Pa, ²²²Pa, ²²³Pa, ²²⁴Pa, ²²⁵Pa, ²²⁶Pa, ²²⁷Pa, ²²⁸Pa, ²²⁹Pa, ²³⁰Pa, ²³¹Pa, ²³²Pa, ²³³Pa, ²³⁴Pa, ²³⁵Pa, ²³⁶Pa, ²³⁷Pa, ²³⁸Pa.
- Uranium (Z=92):** ²²⁶U, ²²⁷U, ²²⁸U, ²²⁹U, ²³⁰U, ²³¹U, ²³²U, ²³³U, ²³⁴U, ²³⁵U, ²³⁶U, ²³⁷U, ²³⁸U, ²³⁹U, ²⁴⁰U.
- Neptunium (Z=93):** ²²⁷Np, ²²⁸Np, ²²⁹Np, ²³⁰Np, ²³¹Np, ²³²Np, ²³³Np, ²³⁴Np, ²³⁵Np, ²³⁶Np, ²³⁷Np, ²³⁸Np, ²³⁹Np, ²⁴⁰Np, ²⁴¹Np.
- Plutonium (Z=94):** ²³²Pu, ²³³Pu, ²³⁴Pu, ²³⁵Pu, ²³⁶Pu, ²³⁷Pu, ²³⁸Pu, ²³⁹Pu, ²⁴⁰Pu, ²⁴¹Pu, ²⁴²Pu, ²⁴³Pu.
- Americium (Z=95):** ²³²Am, ²³³Am, ²³⁴Am, ²³⁵Am, ²³⁶Am, ²³⁷Am, ²³⁸Am, ²³⁹Am, ²⁴⁰Am, ²⁴¹Am, ²⁴²Am, ²⁴³Am.

The diagram shows the following decay paths:

- Alpha decay (α):** Indicated by a yellow background.
- Beta decay (β):** Indicated by a blue background.
- Gamma decay (γ):** Indicated by a black background.

The diagram illustrates the complex decay chains of these isotopes, showing how they eventually reach stable or long-lived states.